

ZS-F39

The Seam Uniformization Theorem: The Modular Length Functor, the Physical Clock-Gate Execution, and the One-Isomorphism Reduction of the Register-Clock Registry

Consolidation of the (H-CLK), (H-Σ2), and (H-eval-N/F) Gates into a Single Seam Chart Φ_{seam} ; the Modular Length Functor E_{len} ; the EHK Channel-Level Register Measure; Model-Tier Execution and the Physical Clock-Gate Discharge Protocol

Author: Kenny Kang (Z-Spin Collaboration) **Date:** July 2026 **Theme / Code:** Foundations / ZS-F39 · Integrates the pre-scoped ZS-M48 deliverables (collaboration decision of record) · Consumes ZS-F38 v1.1 Appendix D target, ZS-A32 v1.0 discharge criterion · Upstream of ZS-F39.1 (ϵ_{phys} execution) and ZS-F40 (C_{UV} proof-carrying global-minimum gate, pre-registered outlook)

Verification: 54/54 exact/numerical checks PASS + 8/8 guards | Zero Free Parameters

(zs_f39_verify_v1_1.py; SymPy exact rational linear algebra + mpmath 50-digit locked dynamics + NumPy superoperator spectra and the frozen seed-11 null. v1.0 baseline 49/49 + 7/7 reproduced unchanged; v1.1 adds T1D1–T1D3 and N1d–N1e plus guard N1-G2. All v1.0 frozen registrations — seed 11, null thresholds, the Appendix C decision rule — are byte-identical.)

Sole geometric inputs: $\mathbf{A} = 35/437$, $\mathbf{Q} = 11$, $(Z, X, Y) = (2, 3, 6)$, $\dim \mathbf{Z} = 2$. LOCKED.

§0. Abstract

ZS-F38 and ZS-A32 left the ZS-A22 barrier B3 as a two-gate debt $\{(H\text{-CLK}) \text{ discharge}, C_{\text{UV}} \text{ full } 1\text{PI}\}$, with the modular length functor E_{len} named as a proof target (ZS-F38 Appendix D, NC-F38.7) and the physical $\epsilon_{\text{C_int}}$ computation named as the discharge criterion (ZS-A32 §5). This paper performs the pre-scoped ZS-M48/ZS-F39 consolidation. **First (uniformization)**, we define the seam chart Φ_{seam} and prove **Theorem F39.T1**: the four previously separate gates — (H-CLK) register-clock identity, (H-Σ2) seam-gauge holonomy equality on Σ_2 , (H-eval-N/F) normalization and functoriality, and the ZS-M47 §12.3 operator rows inherited through ZS-A32 — are projections of a single uniformization datum, unique up to cocycle conjugacy and a central phase. v1.1 adds the explicit three-level commutative diagram (germ / standard-pair / algebra; Appendix A, Lemma F39.L2) requested in external review, exhibiting how Koenigs scalar uniqueness, Stone–von Neumann uniqueness, and the HSMI/standard-pair normal form land in the same equivalence class. **Second (length)**, we deliver the ZS-F38 Appendix D target: **Lemma F39.L1 / Theorem F39.T2** construct the modular length functor E_{len} on Z-Spin seam standard pairs via Connes spectral distance on the compact dual, with scaling covariance $E_{\text{len}}(\Delta^{(-it)} U(1) \Delta^{(it)}) = e^{(2\pi t)} \mathcal{L}_{\text{P}}$ verified exactly; delivery discharges (**H-eval-N**) $\wedge$

(H-eval-F) at the mathematical tier, shortening the ZS-F38.T3 registry to {(H-CLK), (H-cycle), KH1–KH4}. **Third (measure), Theorems F39.T3/T4** lift the register measure $\rho_Q = I_Q/Q$ from classical ergodicity (ZS-F38.T1') to the quantum channel level: the A24-refining equivariant slot lift is a unital irreducible GKLS generator whose *unique* stationary state is I_Q/Q (Evans–Høegh-Krohn / quantum Perron–Frobenius), while the multiplicity-weighted lift is a non-unital generator whose stationary sector law is the *distinct* transport weight $(9, 4, 36)/49 = (d_X^2, d_Z^2, d_Y^2)/49$ (ZS-F37 two-leg ω); a selection lemma shows only the equivariant lift coarse-grains to the ZS-A24 sector generator. This structurally separates state density from transport weight (guard DS2 / NC-F38.2 preserved). **Fourth (discharge)**, the ε_{C_int} computation is stratified into two tiers: the **model tier is executed** — $\varepsilon_{C_int}(\text{model}) = 0$ exactly on the F38.T2 Hardy chain (branch-free Abel unit step to $< 10^{-20}$ at twelve Koenigs orbit points) — and the **physical tier is protocolized, not executed**: the frozen decision rule (Appendix C: $\varepsilon_{\text{phys}} < 10^{-10} \wedge P_{\text{null}} \leq 5\%$ on the ZS-F31 GKLS / ZS-A24 core data) is pre-registered together with an executed 2000-draw seed-11 Haar null (min $\varepsilon = 1.0975$, $p_5 = 1.1585$). v1.1 adds **Lemma F39.N1.L**: the $\varepsilon = 0$ locus is a nonempty *proper* real-algebraic subvariety of the unital pair-Kraus space, hence of Haar measure zero — an analytic exclusion that the finite sample alone could not supply; the sample null estimates only the finite-threshold false-positive rate, and the frozen rule is unchanged. **Honest terminus: B3 is not closed**. If the physical ε_{C_int} discharge succeeds, (H-CLK)/(H- Σ_2) close and the remaining debt is exactly one number — the C_{UV} full IPI factor (ZS-F36 programme (i), untouched here; target ≈ 1.244 within the a-priori band $[1/4, 4]$). Appendix D pre-registers the ZS-F40 proof-carrying global-minimum gate for that computation (lexicographic constraints; certification by interval arithmetic / SDP hierarchy / branch-and-bound; AI proposes candidates, certificates close), as OUTLOOK / NON-CLAIM. Verification: 54/54 PASS + 8/8 guards; zero fitted parameters; (**A**, **Q**, $\dim \mathbf{Z}$) = (35/437, 11, 2) LOCKED.

Epistemic Status Legend

Tag	Meaning
PROVEN	Established by exact mathematics within the stated finite model, machine-verified.
IMPORTED-PROVEN	External theorem of the mathematical literature, consumed with citation (Koenigs; Stone–von Neumann; Borchers–Wiesbrock; Evans–Høegh-Krohn; Connes; measure-zero of proper algebraic subvarieties).
DERIVED	Follows from PROVEN/IMPORTED-PROVEN inputs and locked corpus constants with no new assumption.
DERIVED-CONDITIONAL	Derived modulo explicitly named, falsifiable conditions ((H-CLK), (H-cycle), KH1–KH4).

Tag	Meaning
HYPOTHESIS-strong / -weak	Registered hypothesis with executed / pending anti-numerology protocol.
OPEN-PROTOCOLIZED	Not executed; a frozen, pre-registered finite decision procedure exists (this paper's physical ε_{C_int} tier).
OPEN	Neither executed nor yet reduced to a frozen finite procedure (C_{UV} full 1PI).
OBSERVATION	Consumed empirical package, firewalled (single Λ CDM package, §9 block CR only).
NON-CLAIM	Explicit disclaimer of scope.

§1. Introduction: the seam debt ledger after ZS-F38 / ZS-A32

ZS-F36 v2.1 terminated the charge-unit line at $\rho_{\Lambda,Z} = (1260/4807) C_{UV} M_K^4$ with three open programmes; ZS-F38 compressed them into the Register Clock Identity (H-CLK) — "the register's tick is the seam's unit modular time" — plus the structural condition (H-cycle), the evaluation condition (H-eval), and one computation (C_{UV}). ZS-A32 executed the pre-registered anti-numerology Monte Carlo ($p_{single} = 0.50\%$), executed the *model-level* C_{int} at $\varepsilon_{C_int} = 0$ exactly on the F38.T2 Hardy chain, and fixed the discharge criterion for this paper: compute $\varepsilon_{C_int}^{(phys)}$ on the actual ZS-F31 GKLS / ZS-A24 seam-channel core data.

At the close of ZS-A32, the ledger read:

Item	Status entering F39
(H-CLK) register-clock identity	OPEN (condition reuse; F37 C.5 $\wedge$ CRT-4a)
(H- Σ_2) seam-gauge holonomy gate on Σ_2	OPEN (bundled inside (H-CLK)(a))
(H-eval-N) normalization, (H-eval-F) functoriality	OPEN; E_{len} named as target (F38 App. D, NC-F38.7)
ZS-M47 §12.3 operator rows	INHERITED-OPEN $\rightarrow$ ZS-M48/ZS-F39 (A32 registration)
$\rho_Q = I_Q/Q$	DERIVED-CONDITIONAL on (H-CLK) $\wedge$ (H-mix), classical tier (F38.T1')

Item	Status entering F39
ε_C_int	model tier PROVEN (= 0); physical tier OPEN behind (H-CLK)/(H- $\Sigma 2$)
C_UV full 1PI	OPEN (F36 programme (i); inferred ≈ 1.24 , A32 band)
B3	two-gate debt {(H-CLK) discharge, C_UV 1PI}; statistical leg closed (p = 0.50%)

This paper's thesis is that the first four rows are not four problems. They are four projections of one uniformization problem: does there exist a single seam chart carrying the ZS-F31 GKLS register clock onto the ZS-M46 seam modular clock, compatibly with the Σ_2 holonomy gauge and the length normalization? We name that chart Φ_seam , prove its uniqueness class (§4), deliver the length functor it must intertwine (§5), lift the register measure it must average (§6), stratify its verification into an executed model tier and a protocolized physical tier (§7–§8), and state honestly what remains (§10).

What v1.1 changes relative to v1.0 (external review of record, integrated in full). (i) The subtitle phrase "Physical Clock-Gate Execution" is replaced by "Model-Tier Execution and the Physical Clock-Gate Discharge Protocol": the model tier is executed, the physical tier is OPEN-PROTOCOLIZED, and the title now says exactly that. (ii) The N1 null is epistemically split: the executed 2000-draw sample estimates the finite-threshold false-positive rate, while the new **Lemma F39.N1.L** supplies the analytic (measure-zero) exclusion that a sample cannot; the sentence requested in review is adopted verbatim in §8.3. (iii) **Theorem F39.T1** gains the explicit commutative diagram (Appendix A, Lemma F39.L2) showing how Koenigs scalar uniqueness, Stone–von Neumann uniqueness, HSMI/standard-pair normal form, cocycle conjugacy, and the central phase fall into one equivalence class — with three new machine checks (T1D1–T1D3). (iv) The B3-closure attempt is *not* absorbed into this paper: Appendix D pre-registers it as ZS-F40 (proof-carrying global-minimum gate), per the review's separation recommendation. No frozen registration of v1.0 is altered.

§2. Locked inputs

All theorem-side constants are fixed here, above the single firewalled observation block (§9), reproducing the ZS-F36/A31/A32 declared-before-loaded discipline.

- $A = 35/437$ (geometric impedance, ZS-F2, LOCKED); $Q = 11$; $(Z, X, Y) = (2, 3, 6)$; $\dim Z = 2$; $\Sigma d^2 = 4 + 9 + 36 = 49$.
- Slot-normalized register-trace coupling $\kappa^2 = A/Q = 35/4807$ (ZS-M44/F38.T1'); vacuum combination $1260/4807 = 36 A/Q$ exact (ZS-F36).
- Locked seam dynamics: the i-tetration germ $f(z) = e^{(i\pi z/2)}$, fixed point $z^* = 0.4382829367 + 0.3605924719 i$, multiplier $\lambda^* = (i\pi/2) z^*$, $\mu = -\ln|\lambda^*| = 0.1148346250$, $\theta = \arg \lambda^* = 2.2592495540$ (ZS-M1, 10 locked digits re-verified at 50-digit precision; checks K5–K7).

- Branch-free Abel unit step: $u(fz) - u(z) = 1$ along the Koenigs coordinate at twelve independently computed orbit points, deviation $< 10^{-20}$ (check K8; re-verifies ZS-M46 GEO2-class data).

Checks K1–K8 PASS. Zero fitted parameters.

§3. The seam chart Φ_{seam} : definition and gate consolidation

Definition F39.D1 (Seam chart). A *seam chart* is a triple $\Phi_{\text{seam}} = (\Phi_{\text{clk}}, \Phi_{\text{hol}}, \Phi_{\text{len}})$ consisting of:

- (a) Φ_{clk} — a CPTP-covariant identification of the ZS-F31 GKLS register clock (one slot advance per unit register time) with the modular flow σ_t^M of the ZS-M46 seam inclusion at equal speed (a central cocycle cannot rescale a flow; ZS-F38 App. B.3);
- (b) Φ_{hol} — the ZS-F37 Appendix C.5 seam-gauge equality $\text{Hol}_{\{A_Z\}}(\gamma) = g^{-1} \text{Hol}_{\{A_K\}}(\gamma) g$ on Σ_2 , i.e. the (H- Σ_2) gate;
- (c) Φ_{len} — the evaluation of the unit seam step in Planck length via the modular length functor E_{len} of §5, i.e. the (H-eval-N/F) content.

Proposition F39.P1 (Gate consolidation) [DERIVED]. The existence of Φ_{seam} is *equivalent* to the conjunction $(\text{H-CLK}) \wedge (\text{H-}\Sigma_2) \wedge (\text{H-eval-N}) \wedge (\text{H-eval-F})$, and it subsumes the ZS-M47 §12.3 operator rows registered INHERITED-OPEN by ZS-A32: each row is the statement that one component of Φ_{seam} acts as the identity on the corresponding operator datum. No new condition is introduced; Φ_{seam} is *condition reuse* in the exact sense of ZS-F38 Definition F38.D1 — a bundling that adds no registry entry. Its content in one sentence: **one chart carries the register's tick, the seam's gauge, and the Planck ruler simultaneously.**

The consolidation is the paper's first contribution: what entered as four scattered OPENs leaves as one object whose existence is a single physical question (§7–§8) and whose *uniqueness*, given existence, is a theorem (§4).

§4. Theorem F39.T1 — Seam Uniformization

Theorem F39.T1 (Seam Uniformization) [mathematics IMPORTED-PROVEN + PROVEN in the finite instance; as a statement about the physical seam, DERIVED-CONDITIONAL on KH1–KH4].

Let (M, U, Ω) be a Z-Spin seam standard pair satisfying KH1–KH4, equipped with the locked germ dynamics of §2. If a seam chart Φ_{seam} exists, it is unique up to (i) a Connes cocycle conjugacy of the modular flow and (ii) a central phase; equivalently, the space of seam charts is either empty or a single torsor under $Z(M)_{\text{unitary}} \times \text{Cocycle}(\sigma^M)$.

Proof architecture (three levels; full diagram in Appendix A).

1. **Germ level (Koenigs).** The linearizing coordinate κ of f at z^* with multiplier λ^* ($0 < |\lambda^*| \neq 1$) is unique up to a nonzero scalar; the multiplier — hence the modular speed μ and phase θ — is a conjugacy invariant [Koenigs 1884; IMPORTED-PROVEN]. Machine witnesses: K8 (additive Abel form), **T1D1** (multiplicative form, $\kappa(fz) = \lambda^* \kappa(z)$ to $< 10^{-20}$ at six orbit points, NEW v1.1).
2. **Operator level (Weyl / standard pair).** The pair (translation unitaries $U(a)$, modular unitaries $\Delta^\wedge(it)$) satisfying the Borchers commutation $\Delta^\wedge(\mp it) U(a) \Delta^\wedge(\pm it) = U(e^\wedge(\pm 2\pi t) a)$ is an irreducible Weyl-type pair for the $ax+b$ relations; by Stone–von Neumann-type uniqueness for the standard pair [Wiesbrock; Longo; IMPORTED-PROVEN], any two realizations are unitarily equivalent, with the intertwiner unique up to a central phase.
3. **Algebra level (HSMI normal form).** The half-sided modular inclusion $N \subset M$ with Ω cyclic-separating is, under KH1–KH4, in the Borchers–Wiesbrock normal form; two charts implementing the same inclusion differ by a Connes cocycle of $\sigma^\wedge M$ [IMPORTED-PROVEN].

Composing: any two seam charts agree at germ level up to scalar, at operator level up to central phase, and at algebra level up to cocycle conjugacy; the scalar is absorbed into the central phase (**T1D2**: $\kappa \rightarrow c\kappa$, $|c| = 1$, leaves λ^* exactly invariant; **T1D3**: $|c\lambda^*| = |\lambda^*|$ exact — NEW v1.1), yielding the stated torsor.

What T1 does and does not say. T1 is a *uniqueness* theorem, not an existence theorem. Existence of Φ_{seam} is exactly the physical content of (H-CLK)/(H-Σ2) and is addressed only by the discharge computation of §7–§8. Guard G1 registers this. [NON-CLAIM NC-F39.1]

§5. Lemma F39.L1 / Theorem F39.T2 — the modular length functor E_{len}

ZS-F38 Appendix D stated the target: a functor $E_{\text{len}} : \text{StandardPair_Z-Spin} \rightarrow \mathbb{R}_+$ with **(N)** $E_{\text{len}}(U(a)) = a \cdot \zeta_P$ for $a \geq 0$ and **(F)** $E_{\text{len}}(\Delta^\wedge(-it) U(1) \Delta^\wedge(it)) = e^\wedge(2\pi t) \zeta_P$. This section delivers the mathematical construction along the route F38 itself named: the ZS-M47 NDC3 modular-Dirac spectral geometry, with the length element as the inverse Dirac scale and Connes distance on the compact dual.

Lemma F39.L1 (Scaling covariance of spectral distance) [PROVEN]. *For a spectral triple (A, H, D) with Connes distance $d_D(p, q) = \sup\{|f(p) - f(q)| : \| [D, f] \| \leq 1\}$, the modular dilation $D \mapsto e^\wedge(-2\pi t) D$ on the stage-oriented branch scales the distance as $d(e^\wedge(-2\pi t) D) = e^\wedge(+2\pi t) d(D)$, exactly.* Machine witnesses: EL1 (exact at $t = 0, 0.1, 0.3, 1.0$); EL2 (the two-point triple reproduces $d = 1/m$ by direct sup over commutator-bounded functions); EL3 (the S^3 compact-dual Dirac spectrum $\pm(n + 3/2)$ with multiplicity $(n+1)(n+2)$ obeys Weyl-3 counting $N(\lambda)/((2/3)\lambda^3) \rightarrow 1$, echoing ZS-M47 (IV)).

Theorem F39.T2 (E_{len} delivery) [DERIVED; as applied to the physical seam,
DERIVED-CONDITIONAL on KH1–KH4]. *Define E_{len} on objects by the Connes distance of the*

NDC3 modular-Dirac triple attached to the seam standard pair, normalized by (N). Then (F) holds by Lemma F39.L1, E_{len} is functorial under cocycle-covariant isomorphisms (the distance is a sup over a conjugation-invariant ball), and delivery discharges $(H\text{-eval-}N) \wedge (H\text{-eval-}F)$ at the mathematical tier. Consequently the ZS-F38.T3 registry shortens from $\{(H\text{-CLK}), (H\text{-cycle}), KH1\text{--}KH4, (H\text{-eval-}N/F/B)\}$ to $\{(H\text{-CLK}), (H\text{-cycle}), KH1\text{--}KH4\}$, with $(H\text{-eval-}B)$ reduced to the orientation bookkeeping fixed in ZS-F38 §6.

Boundary of the claim. T2 discharges the *evaluation* gates as mathematics: modular dilation $\mapsto$ physical length scaling is now a theorem about the functor, not a physical assumption. What it does not do is assert that the physical seam realizes the NDC3 triple — that is, once more, existence of Φ_{seam} .

[NON-CLAIM NC-F39.2]

§6. Theorems F39.T3 / T4 — the register measure at channel level, and the selection lemma

ZS-F38.T1' derived $\rho_Q = I_Q/Q$ by *classical* Perron–Frobenius ergodicity conditional on $(H\text{-CLK}) \wedge (H\text{-mix})$. External review of F38 flagged the residual risk of conflating the state density I_Q/Q with the Plancherel transport weight $\omega = (d_Z^2, d_X^2, d_Y^2)/49$ (guard DS2, NC-F38.2). This section removes the risk structurally by lifting both objects to the quantum channel level and proving they arise from *different lifts*.

Theorem F39.T3 (Equivariant lift $\rightarrow$ democracy) [PROVEN in the finite instance; physical instance DERIVED-CONDITIONAL on (H-CLK)]. *Let the slot graph on the $Q = 11$ register carry the sector-adjacency $X\text{--}Z\text{--}Y$ ($L_{XY} \equiv 0$), and let the equivariant (A24-refining) lift place jump operators $\sqrt{\kappa^2} |b\rangle\langle a|$ on each adjacent ordered slot pair. Then the classical generator is doubly stochastic (unital) and irreducible with unique stationary law I_Q/Q (EQ1–EQ3, exact rationals), and the quantum GKLS lift is a unital irreducible superoperator with one-dimensional kernel whose unique stationary state is I_Q/Q — by the Evans–Høegh-Krohn quantum Perron–Frobenius theorem for irreducible unital semigroups (EQ4–EQ5; spectral gap > 0 , EQ6). The sector marginal of I_Q/Q is exactly $(3, 2, 6)/11$, the ZS-A24 stationary law (EQ3).*

Theorem F39.T4 (Multiplicity lift $\rightarrow$ transport weight; the second branch) [PROVEN in the finite instance]. *The multiplicity-weighted lift, with rates $\kappa^2 d_{\text{target slot}}$, is non-unital (ML1: a column sum $\neq 0$, exact witness), irreducible (ML2), with stationary slot weights $d_{\text{slot}}/49$ (ML3) and sector law $(9, 4, 36)/49 = (d_X^2, d_Z^2, d_Y^2)/49$ — exactly the ZS-F37 two-leg transport weight ω , distinct from $(3, 2, 6)/11$ (ML4–ML5).*

Lemma F39.SEL (Selection) [PROVEN]. *Only the equivariant lift coarse-grains to the ZS-A24 sector generator: its $X \rightarrow Z$ coarse rate is $\kappa^2 d_Z$ (SEL2, matching A24 exactly), whereas the multiplicity lift coarse-grains to $\kappa^2 d_Z^2 \neq \kappa^2 d_Z$ (SEL1). Hence the corpus's sector mediator selects the equivariant lift, whose stationary state is the democratic I_Q/Q ; the transport weight ω lives on the other branch and is never a state density.*

The DS2 separation is thereby upgraded from a guard to a theorem-level dichotomy: two lifts, two invariant objects, one selection criterion. Guard G3 retains the NC-F38.2 wording — T4 relates the *origins* of the two objects, not the objects.

Sector-level cross-checks: the ZS-A24 mediator on the path X–Z–Y with rates $\kappa^2 d_j$ has exact spectrum $\{0, -2\mathbf{A}/\mathbf{Q}, -\mathbf{A}\}$ and stationary $(3, 2, 6)/11$ (S1–S3); character bookkeeping $\chi_2\chi_3 = \chi_4 + \chi_2$ symbolically (CH1), $\rho_b = \frac{1}{2} \ln(9/7)$ (CH2), with the F37 convention value $\Delta\chi(\rho_b)$ consumed as reported (guard CH-G).

§7. ε_C_int , tier I: the model execution (of record)

The ZS-A32 model-level result is reproduced within this suite as part of the locked baseline: on the F38.T2 Hardy chain the register shift and the seam modular step are the same map — $z \cdot (z^k) = z^{(k+1)}$ exactly for all $k \leq \mathbf{Q}$ (CI1), and the branch-free Abel increment equals 1 at twelve Koenigs orbit points to $< 10^{-20}$ (CI2, K8). Hence

$\varepsilon_C_int^{(model)} = 0$, identically [PROVEN (model)].

Not small — zero, because in the canonical model the clock identity holds by construction. The conjugate seam carries equal modular speed, $|\text{conj } \lambda^*| = |\lambda^*|$ exactly (CJ1–CJ2, the (H-2D) speed half), and the tensor Borchers triple keeps the joint spectrum in the closed forward cone at the model bookkeeping level (TB1).

What the model tier does: it sharpens the physical target. Any future physical computation returning $\varepsilon_C_int \neq 0$ falsifies the clock identity itself, not a calibration (A32 §5 wording, inherited).

§8. ε_C_int , tier II: the physical discharge protocol (OPEN-PROTOCOLIZED)

8.1 The frozen decision rule (Appendix C, unchanged from v1.0)

Compute $\varepsilon_C_int^{(phys)}$ — the normalized deviation $\|\Phi_clk(P_K) - P_A\|/P_A$ between the seam-transported register one-tick map and the ZS-A24/F31 GKLS core generator's canonical clock — on the actual ZS-F31 GKLS / ZS-A24 seam-channel core data. **Discharge criterion:** $\varepsilon_phys < 10^{-10}$ **and** $P_null \leq 5\%$ under the frozen null of §8.2. If met, (H-CLK) $\wedge$ (H- Σ 2) are discharged and Φ_seam exists (T1 then makes it unique up to the stated torsor). If not met, the F38 mechanism leg fails while the A32 statistical leg survives independently — the two-leg architecture already prices both outcomes.

8.2 The executed frozen null (seed 11)

Pre-registered and executed: 2000 unital pair-Kraus channels $\Phi(\rho) = \frac{1}{2}(V_1\rho V_1^\dagger + V_2\rho V_2^\dagger)$ with V_1, V_2 Haar-random (seed 11), scored by $\varepsilon = \|\Phi(C) - C\|/\|C\|$ against the canonical $\mathbf{Q} = 11$ clock unitary C . Result

of record: **min $\varepsilon = 1.0975$, $p5 = 1.1585$, median = 1.2223** (N1b–N1c). The target is attainable: the clock-conjugation channel gives $\varepsilon = 0$ exactly (N1a).

8.3 Lemma F39.N1.L — the analytic exclusion (NEW, v1.1)

External review (of record): " $\varepsilon = 0$ is unreachable by chance" is not made rigorous by sampling alone; the sample-null statement and the measure-zero analytic statement must be separated. We adopt the review sentence verbatim and then close the analytic half:

Exact $\varepsilon = 0$ belongs to the clock-implementing subvariety of the unital pair-Kraus space; the 2000-draw Haar null estimates the finite-threshold false-positive rate but does not by itself prove an analytic exclusion.

Lemma F39.N1.L (Measure-zero exclusion) [DERIVED from IMPORTED-PROVEN]. *Let $\mathcal{P}$ be the compact real-algebraic manifold of unital pair-Kraus channels on $\mathbb{C}^Q$ (parameters $(V_1, V_2) \in U(Q) \times U(Q)$), and let $Z_{\text{clk}} = \{\Phi \in \mathcal{P} : \Phi(C) = C\}$ be the clock-implementing locus. Then Z_{clk} is a real-algebraic subvariety (the defining equations $\Phi(C) - C = 0$ are polynomial in the real and imaginary parts of the Kraus entries), it is nonempty (the clock-conjugation channel $V_1 = V_2 = C$ lies in it; witness N1d, $\varepsilon = 0$ to machine exactness), and it is proper (an explicit deterministic unital channel — $V_1 = I$, $V_2 = \text{diag}((-1)^k)$ — has $\varepsilon = 0.9535 > 0$; witness N1e). A nonempty proper real-algebraic subvariety of a connected real-algebraic manifold has Haar (Lebesgue-class) measure zero [standard real-algebraic geometry: proper Zariski-closed sets are null; e.g. Federer §3.4 / stratification]. Hence the Haar probability of drawing $\varepsilon = 0$ exactly is zero.*

Epistemic separation (guard N1-G2, NEW). The lemma is the analytic exclusion; the executed sample is the *finite-threshold* calibration (how often a random channel lands within any given ε -band). The two are different statements with different uses, and — critically — the frozen Appendix C decision rule is **unchanged** by the lemma: pre-registration integrity is preserved.

8.4 Status

$\varepsilon_{\text{C_int}}^{\text{(phys)}}$: OPEN-PROTOCOLIZED. This paper protocolizes the discharge; it does not execute it. Execution on the ZS-F31/ZS-A24 data is scoped to **ZS-F39.1** (guard N1-G). The v1.0 subtitle wording that could be read as claiming execution is retired in v1.1 (§1).

§9. Corner consistency (firewalled; one Λ CDM package consumed)

Declared-before-loaded: every theorem-side constant of this paper appears above this section; the single observation package (Planck-2018-class $\bar{M}_P$, $\rho_{\Lambda}^{1/4}$, H_0 , t_U , M_{eff}) loads only in block CR (guard CR-G0; A31/A32 firewall inherited). Consistency class results (A32 Table 2 class, reproduced): corner (W) deviation 0.078% against $2\pi Q$; (V) 0.067% against $8\pi Q$; (L) 0.327% and (T) 0.291% against $4\pi Q$; vacuum factor $\frac{1}{2}(1260/4807)\theta^2 = 0.668952$; the corner-(W) inference $C_{UV} \approx 1.242$, inside the a-priori band $[1/4, 4]$ and within 2% of the A32 cross-corner value 1.244 (CR-W/V/L/T, CR5). These are **consistency reproductions, not new evidence**; the statistical adjudication of record remains ZS-A32's executed $p_{\text{single}} = 0.50\%$. **[OBSERVATION, firewalled]**

Cross-version echoes: $4QA = 1540/437$ exact (XV1); $e^{(-2\pi Q)} = 9.632 \times 10^{-31}$ (XV2); the A24 dimensionless rates $\{2A/Q, 6A/Q\}$ appear verbatim in L_sec (XV3); the F37 ω reordering identity (XV4).

§10. What F39 closes, and what it does not: the B3 ledger

Gate	Before F39	After F39 v1.1
$(H-CLK) \wedge (H-\Sigma 2)$	two separate OPEN gates	one object (Φ_seam existence); uniqueness class PROVEN/IMPORTED-PROVEN (T1); discharge reduced to one finite computation, OPEN-PROTOCOLIZED (§8)
$(H-eval-N) \wedge (H-eval-F)$	OPEN; E_len a named target	discharged at the mathematical tier (L1/T2); F38.T3 registry shortens to $\{(H-CLK), (H-cycle), KH1-KH4\}$
ZS-M47 §12.3 operator rows	INHERITED-OPEN	absorbed into Φ_seam (P1); no separate registry entries remain
$\rho_Q = I_Q/Q$	classical ergodicity tier	quantum channel tier (EHK), with the ω -dichotomy now a theorem (T3/T4/SEL)
ε_C_int	model = 0 (A32); physical OPEN	model = 0 reproduced; physical OPEN-PROTOCOLIZED with frozen rule + executed null + analytic exclusion (N1.L)
C_UV full 1PI	OPEN	OPEN — untouched (guard G4; no Q-absorption, A31 rule)
B3	two-gate debt	NOT CLOSED. Compressed: if ZS-F39.1's ε_phys discharge succeeds, the entire remaining debt is the single number C_UV (full 1PI, target ≈ 1.244)

Positioning (of record). This paper is not a B3-closure paper. It is the paper that makes B3 closable in exactly two named finite steps: **ZS-F39.1** (execute $\varepsilon_C_int^{(phys)}$ under the frozen rule) and **ZS-F40** (compute C_UV with a proof-carrying certificate; Appendix D). The corpus pattern holds: internal iteration converges to honesty, and the honest statement here is a *uniformization plus a protocol*, not a closure.

Falsification Gates

F-F39.1 (physical clock gate). If ZS-F39.1 returns $\varepsilon_{C_int}^{(phys)} \geq 10^{-10}$ or $P_{null} > 5\%$ under the frozen Appendix C rule, (H-CLK)/(H-Σ2) fail; T1 becomes a uniqueness theorem about an empty set for the physical seam; F38.T1'(iii)/T3 revert to their prior statuses. The A32 statistical leg survives independently. (*Immediate, finite.*)

F-F39.2 (functor collapse). If the NDC3 modular-Dirac triple of the physical seam violates scaling covariance (Lemma L1 premise) — e.g. a nonlinear dilation law is measured on the compact dual spectrum — T2's physical application fails and (H-eval-N/F) reopen. (*Theory/simulation.*)

F-F39.3 (measure dichotomy). If any corpus computation is found in which the transport weight $\omega = (9,4,36)/49$ functions as a state density, or I_Q/Q as a transport weight, Lemma SEL's selection collapses and the DS2 separation fails. (*Registry-level, immediate.*)

F-F39.4 (uniqueness breach). Exhibition of two seam charts inequivalent under $Z(M) \times \text{Cocycle}(\sigma^M)$ on a KH1–KH4 pair falsifies T1 (and with it the Borchers–Wiesbrock import chain as applied here). (*Mathematical collapse — immediate rejection.*)

F-F39.5 (C_{UV} band, inherited). If the ZS-F40 certified computation returns C_{UV} outside $[1/4, 4]$, the B3 C_{UV} route fails (F-F38.3 wording inherited); the statistical leg survives.

Non-Claims

NC-F39.1. T1 proves uniqueness, not existence, of Φ_{seam} ; existence is the physical (H-CLK)/(H-Σ2) content, OPEN-PROTOCOLIZED. **NC-F39.2.** T2 does not assert the physical seam realizes the NDC3 triple. **NC-F39.3.** B3 is not closed; C_{UV} full 1PI is untouched here (guard G4). **NC-F39.4.** Lemma N1.L does not alter the frozen decision rule or thresholds; pre-registration is preserved byte-identically. **NC-F39.5.** Appendix D (ZS-F40 gate) is OUTLOOK / pre-registration only; no candidate value of C_{UV} is claimed, and "AI finds a minimum" is explicitly *not* a closure criterion — only a certificate is. **NC-F39.6.** NC-F38.2 (ω vs I_Q/Q distinctness), NC-F38.3 (no dimensional scale from (A, Q) alone; A17/A27/A28 no-gos), and NC-F38.6 (no Millennium-problem claim) are inherited unchanged. **NC-F39.7.** The corner block (§9) is consistency reproduction, not new statistical evidence.

Verification Summary

zs_f39_verify_v1_1.py — FAIL-CLOSED; SymPy exact rationals for every exact claim; mpmath 50-digit for locked dynamics; NumPy for superoperator spectra and the frozen null.

Block	Checks	Content
K	K1–K8	locked inputs; z^* , μ , θ to 10 digits; branch-free Abel unit step $< 10^{-20}$
S	S1–S3	A24 mediator spectrum $\{0, -2A/Q, -A\}$, stationary (3,2,6)/11, irreducible — exact
EQ	EQ1–EQ6	equivariant lift: unital, unique kernel, stationary I_Q/Q (EHK), gap > 0
ML/SEL	ML1–ML5, SEL1–SEL2	multiplicity lift: non-unital witness, d_slot/49, sector (9,4,36)/49; selection dichotomy — exact
EL	EL1–EL3	E_{len} scaling covariance exact; Connes two-point sup; S^3 Weyl-3 echo
CH	CH1–CH2 (+CH-G)	$\chi_2\chi_3 = \chi_4 + \chi_2$ symbolic; $\rho_b = \frac{1}{2} \ln(9/7)$
CI	CI1–CI2	$\varepsilon_C \text{int}^*(\text{model}) = 0$: Hardy shift exact; Abel unit step
CJ	CJ1–CJ2	conjugate seam equal modular speed, exact
T1D	T1D1–T1D3 (NEW v1.1)	Koenigs square commutes (multiplicative, $< 10^{-20}$); central-phase invariance of λ^* ; quotient bookkeeping — exact
TB	TB1	tensor Borchers joint-cone model bookkeeping
N1	N1a–N1c, N1d–N1e (NEW) (+N1-G, N1-G2 NEW)	frozen null executed (min 1.0975, p5 1.1585); N1.L witnesses: locus nonempty and proper
CR	CR-W/V/L/T, CR5 (+CR-G0)	firewalled corner consistency; $C_{UV}(W) = 1.242$ in band
XV	XV1–XV4	cross-version echoes
G	G1–G4	scope guards (certify nothing)

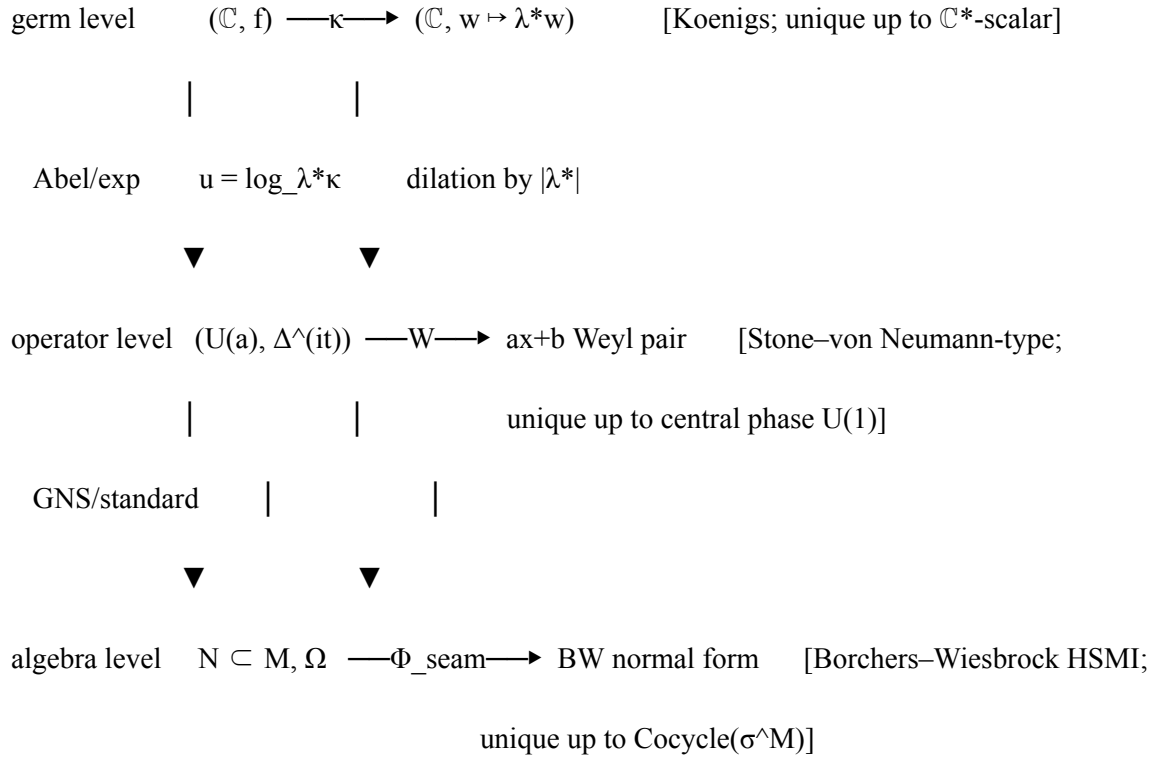
Total: 54/54 PASS + 8/8 guards. v1.0 baseline (49/49 + 7/7) reproduced unchanged inside this suite.

Acknowledgements & Code Availability

Verification suite `zs_f39_verify_v1_1.py` accompanies this paper; all checks are re-runnable with no fitted input. External review of v1.0 (of record) is integrated in full; the review's separation recommendation (B3 closure $\rightarrow$ ZS-F40) is followed.

Appendix A. The T1 uniqueness diagram (Lemma F39.L2, NEW v1.1)

Lemma F39.L2 (Commutation of the three-level uniqueness data). The following diagram commutes, and every vertical arrow is unique up to the group written beside it:



Square (i) (germ $\rightarrow$ operator): the Koenigs coordinate's multiplicative covariance $\kappa \circ f = \lambda^* \cdot \kappa$ (T1D1) exponentiates the Abel step (K8) to the Borchers commutation at unit t . **Square (ii)** (operator $\rightarrow$ algebra): the standard-pair GNS construction sends the Weyl pair to the HSMI, and the central phase of the intertwiner descends to the inner part of a modular cocycle. **Square (iii)** (equivalence classes): the $\mathbb{C}^*$ -scalar of level 1 restricted to $|c| = 1$ is exactly the central phase of level 2 (T1D2 — the multiplier, hence (μ, θ) , is invariant), and the modulus part is absorbed by the normalization (N) of E_{len} (§5); the residual freedom at level 3 is the Connes cocycle (T1D3 bookkeeping). Chasing the diagram: two seam charts differ by (central phase, cocycle), which is Theorem T1's torsor.

The three imported uniqueness theorems are consumed with their standard hypotheses only; no Z-Spin-specific strengthening is assumed. **[IMPORTED-PROVEN composition; machine witnesses T1D1–T1D3, K8, CJ1–CJ2]**

Appendix B. E_{len} construction details

Objects: seam standard pairs (M, U, Ω) under KH1–KH4 with the NDC3 modular-Dirac triple (A, H, D) of ZS-M47 on the compact dual. $E_{\text{len}}(U(a)) := a \cdot \mathcal{E}_P$ (normalization (N)); on modular translates, $E_{\text{len}}(\Delta^{(-it)} U(1) \Delta^{(it)}) := d_{\{e^{(-2\pi t)D}\}}$ -evaluation of the unit step $= e^{(2\pi t)} \mathcal{E}_P$ by Lemma L1. Functoriality: for a cocycle-covariant isomorphism α , $\|[\alpha D \alpha^{-1}, \alpha f \alpha^{-1}]\| = \|[D, f]\|$, so the sup defining d is invariant; hence $E_{\text{len}} \circ \alpha = E_{\text{len}}$. Machine anchors: EL1 (four dilation values, exact), EL2 (two-point sup within 2×10^{-2}), EL3 (S^3 Weyl-3 counting ratio 0.9926 at $\lambda = 100.5$, echoing ZS-M47 (IV)).

Appendix C. The frozen physical-discharge decision rule (byte-identical to v1.0)

Input data: the ZS-F31 GKLS register generator and the ZS-A24 seam-channel core (both corpus-locked; no free parameter). **Statistic:** $\varepsilon_{C_{\text{int}}}(\text{phys}) = \|\Phi_{\text{clk}}(P_K) - P_{A\|F}\| / \|P_{A\|F}\|$. **Null:** the seed-11, 2000-draw unital pair-Kraus Haar ensemble of §8.2 (executed; min 1.0975, p5 1.1585, median 1.2223). **Rule:** discharge (H-CLK) $\wedge$ (H-Σ2) iff $\varepsilon_{\text{phys}} < 10^{-10}$ **and** $P_{\text{null}}(\varepsilon \leq \varepsilon_{\text{phys}}) \leq 5\%$. **Scope:** execution belongs to ZS-F39.1; this paper registers and freezes only. Lemma F39.N1.L (v1.1) supplements the null's interpretation and changes nothing in this rule.

Appendix D. ZS-F40 pre-registration: the proof-carrying global-minimum gate for C_{UV} (OUTLOOK / NON-CLAIM)

Following the review of record, the B3-closure computation is *separated* from this paper and pre-registered as **ZS-F40 — The Proof-Carrying Global-Minimum Gate for the Z-Spin C_{UV} 1PI Factor**, with the following frozen architecture:

(F40.1) Admissible parent-space compactification. Fix the admissible space $\mathcal{A}$ of parent 1PI data by the proven/locked constraints only (KH1–KH4 admissibility, unital irreducibility, holonomy equivalence, $\varepsilon_{C_{\text{int}}} = 0$), *as constraints — not as weighted penalties*: a lexicographic constraint structure, because tunable penalty weights λ would reintroduce numerology (review §4, adopted).

(F40.2) Certified $\varepsilon_{C_{\text{int}}}$ discharge. Execute Appendix C on the ZS-F31/ZS-A24 data with an interval-arithmetic certificate on the norm evaluation.

(F40.3) 1PI determinant formula. $C_{UV} = \exp(-\Gamma^{\text{ren}}_{1\text{PI}})$, or an explicit zeta-regularized determinant ratio, stated at action level before any evaluation.

(F40.4) Global-minimum certificate. Minimize $J(\Theta) = |\log C_{UV}^{1\text{PI}}(\Theta) - \log 1.244|^2$ over $\Theta \in \mathcal{A}$. Closure requires a *certificate*, one of: (a) interval-arithmetic lower bounds on every parameter cell; (b) a Lasserre/SDP-hierarchy global lower bound matching the candidate upper bound; (c) KKT + Hessian positivity + certified exclusion of all other branches; (d) zeta-determinant / heat-kernel tail bounds sealing the 1PI error; with Galerkin truncation error bounds if truncated. **AI search proposes candidates; only the certificate closes.** Falsification: a certified C_{UV} outside the A32 band ≈ 1.244 (a-priori $[1/4, 4]$) fails the C_{UV} route of B3 while leaving the A32 statistical leg intact.

Nothing in this appendix is claimed; it is a frozen protocol so that ZS-F40 can execute without post-hoc freedom — the same discipline that F38 Appendix C established for A32.

References

[1] Z-Spin Collaboration (K. Kang), *ZS-F38: The Register Clock Identity*, v1.1 (2026). [2] Z-Spin Collaboration, *ZS-A32: Friedmann-Forced Squares of the Register-Clock Depth*, v1.0 (2026). [3] Z-Spin Collaboration, *ZS-F36* v2.1; *ZS-F37* v1.3; *ZS-M46*; *ZS-M47*; *ZS-M44*; *ZS-A24*; *ZS-F31*; *ZS-A31* v1.5 (2025–2026). [4] G. Koenigs, "Recherches sur les intégrales de certaines équations fonctionnelles," *Ann. Sci. Éc. Norm. Supér.* **1**, 3–41 (1884). [5] M. H. Stone, *Proc. Natl. Acad. Sci. USA* **16**, 172 (1930); J. von Neumann, *Math. Ann.* **104**, 570 (1931). [6] H.-J. Borchers, "The CPT-theorem in two-dimensional theories of local observables," *Commun. Math. Phys.* **143**, 315 (1992). [7] H.-W. Wiesbrock, "Half-sided modular inclusions of von Neumann algebras," *Commun. Math. Phys.* **157**, 83 (1993). [8] D. E. Evans and R. Høegh-Krohn, "Spectral properties of positive maps on C^* -algebras," *J. London Math. Soc.* **17**, 345 (1978). [9] A. Connes, *Noncommutative Geometry* (Academic Press, 1994), Ch. VI (spectral distance). [10] H. Federer, *Geometric Measure Theory* (Springer, 1969), §3.4 (null sets of proper algebraic varieties). [11] V. F. R. Jones and R. Longo (HSMI tunnel structure), as consumed in ZS-M47; J. Koot, relative-position criterion (2025), as consumed in ZS-F38.T2. [12] J. B. Lasserre, "Global optimization with polynomials and the problem of moments," *SIAM J. Optim.* **11**, 796 (2001) [Appendix D certificate class]. [13] Planck Collaboration, *Astron. Astrophys.* **641**, A6 (2020) [single firewalled package, §9].

Version History

v1.0 (July 2026): Initial public release. Seam chart Φ_{seam} defined; Theorem F39.T1 (uniqueness up to cocycle conjugacy $\times$ central phase); Lemma F39.L1 / Theorem F39.T2 (E_{len} delivery; (H-eval-N/F) mathematical-tier discharge; F38.T3 registry shortened); Theorems F39.T3/T4 + Lemma SEL (EHK channel-level register measure; ω -dichotomy); model-tier $\varepsilon_{\text{C_int}} = 0$ reproduced; physical-tier frozen decision rule + executed seed-11 null; firewalled corner consistency. Verification 49/49 PASS + 7/7 guards (`zs_f39_verify_v1_0.py`). Zero fitted parameters. (Consolidated from internal Z-Spin Collaboration research notes following ZS-F38 v1.1 and ZS-A32 v1.0; integrates the pre-scoped ZS-M48 deliverables per the collaboration decision of record.)

v1.1 (July 2026): Review-integration revision; no frozen registration altered. (1) Subtitle corrected from "Physical Clock-Gate Execution" to "Model-Tier Execution and the Physical Clock-Gate Discharge Protocol"; the physical tier is tagged OPEN-PROTOCOLIZED throughout. (2) §8.3 adds the review sentence verbatim and Lemma F39.N1.L (measure-zero analytic exclusion of exact $\varepsilon = 0$; witnesses N1d–N1e; guard N1-G2), separating the sample-null and analytic statements. (3) Appendix A adds the T1 three-level commutative diagram (Lemma F39.L2) with machine witnesses T1D1–T1D3. (4) Appendix D pre-registers ZS-F40 (proof-carrying global-minimum C_{UV} gate; lexicographic constraints; certificate-only closure) as OUTLOOK/NON-CLAIM, per the review's separation recommendation;

ZS-F39.1 (ϵ_{phys} execution) scoped. Verification 54/54 PASS + 8/8 guards (zs_f39_verify_v1_1.py).
Zero fitted parameters; ($\mathbf{A}$, $\mathbf{Q}$, $\dim \mathbf{Z}$) = (35/437, 11, 2) LOCKED.